

Full Length Article

Hybrid feature-based collaborative disambiguation for multi-instance partial-label learning

Zhen Zhu^{a,b,*}, Yining Sun^{a,c}, Yuanbo Wang^{a,c}, Bingxin Zhou^{a,c}, Yiwen Hu^a, YuXing Zhang^c^a KEYI College, Zhejiang Sci-tech University, Hangzhou, Zhejiang, China^b School of Computer Science & Technology, Beijing Jiaotong University, Beijing, China^c AI lab, Yixiao Technology Co., Ltd., Hangzhou, Zhejiang, China

ARTICLE INFO

Keywords:

Multi-instance learning
 Partial-label learning
 Hybrid feature
 Dual disambiguation
 Dynamic instance filtering

ABSTRACT

Multi-instance partial-label learning (MIPL) deals with the task of learning from multi-instance bags associated with candidate label sets. Each bag contains a set of instances, and only one true label lies within its candidate label set. Existing MIPL methods, such as DEMIPL and MIPLGP, primarily rely on bag-level aggregated features for training and disambiguation. However, the aggregation operation inevitably discards critical instance-level details and reduces feature information density, which limits disambiguation performance. To tackle this issue, we propose a novel MIPL method named **HFC-MIPL** (Hybrid Feature-based Collaborative Disambiguation for MIPL). We first build multi-granular hybrid features by fusing purified instance representations and aligned bag embeddings to retain both local and global semantics. A dual disambiguation scheme combines focusing regularization for distribution shaping with manifold propagation for pseudo-label refinement through feature similarity. In addition, an epoch-adaptive filtering strategy based on raw sigmoid attention determines the purified instance summary used for graph construction. Experiments on image, audio, and medical datasets show that HFC-MIPL achieves competitive performance against the compared MIPL and PLL baselines across different candidate-label ambiguity settings. Code: <https://github.com/hzzhuzhen/HFC-MIPL>.

1. Introduction

In practical machine learning scenarios, due to the high cost of data annotation and the ambiguity in the annotation process (Burns et al., 2025; Dong et al., 2021; Taye, 2023), the application of fully supervised learning is greatly limited. Weak supervision learning has become an important approach to solve this problem (Lin et al., 2024; Witmer & Bhanu, 2024). Multi-Instance Partial Label Learning (MIPL) (Zhou, 2018) combines the characteristics of Multi-Instance Learning (MIL) and Partial Label Learning (PLL) (Ilse et al., 2018; Qiao et al., 2025). Its core setting is: the training data consists of multiple multi-instance bags, each bag contains several instances, and each bag is only assigned a candidate label set with a unique true label hidden in it. The label information of individual instances within the bag is unknown.

Multi-instance partial-label learning (MIPL) exhibits substantial practical value across a diverse range of computer vision and pattern recognition tasks. A prominent application lies in pathological image analysis, where a single pathological slice is modeled as a multi-instance bag and individual cellular regions within the slice serve as instances (Tang et al., 2023); due to the high cost and subjectivity of

fine-grained cellular annotation, clinicians can only provide slice-level candidate lesion labels rather than precise instance-level annotations, a setting that aligns perfectly with the MIPL paradigm. Analogous scenarios exist in video classification (treating videos as bags and frames as instances) and bioacoustic recognition (e.g., bird song classification, with audio clips as bags and sub-segments as instances), where annotators are only able to assign coarse-grained candidate label sets for the entire bag instead of accurate labels for each constituent instance. Such label ambiguity and instance-level annotation scarcity are pervasive challenges in real-world machine learning applications, underscoring the critical need for robust MIPL methodologies to address these practical constraints and unlock the potential of weakly supervised learning in real-world scenarios.

In recent years, a series of multi-instance partial-label learning (MIPL) algorithms including MIPLGP (Tang et al., 2024c), DEMIPL (Tang et al., 2023), ELIMIPL (Tang et al., 2024b) and MIPLMA (Tang et al., 2024a) have been proposed to resolve dual ambiguity over instances and labels: MIPLGP adopts Gaussian process modeling for ambiguous bag data, DEMIPL leverages disambiguation attention embedding, ELIMIPL utilizes conjugate label information with scaled

* Corresponding author.

E-mail addresses: hzzhuzhen@yeah.net (Z. Zhu), 2114944317@qq.com (Y. Sun), 2821479595@qq.com (Y. Wang), 2217784473@qq.com (B. Zhou), 3529730239@qq.com (Y. Hu), 3212191387@qq.com (Y. Zhang).

<https://doi.org/10.1016/j.neunet.2026.109648>

Received 10 May 2026; Received in revised form 21 August 2026; Accepted 16 September 2026

Available online 21 September 2026

0893-6080/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

additive attention, and MIPLMA introduces margin-adjusted loss to mitigate weak-supervision label bias. Although existing MIPL studies have explored attention aggregation and instance-bag information, the reviewed methods provide limited integration of a dynamically purified instance summary with a global bag representation for graph-based label refinement and collaborative optimization. To address this less-explored integration, we propose HFC-MIPL with a hybrid representation, dynamic noise filtering, and a dual-disambiguation collaborative training framework. The contributions are summarized as follows:

- We construct a hybrid feature representation that combines a purified instance summary with an aligned bag embedding before graph construction.
- We construct a unified collaborative training framework equipped with focusing and manifold dual disambiguation to jointly optimize bag-instance losses for resolving dual ambiguity of MIPL.
- We propose an epoch-adaptive noisy instance filtering strategy driven by raw attention scores to suppress low-quality sample interference iteratively.

2. Related work

2.1. Multi-instance learning

Multi-instance learning solves instance-space ambiguity where only bag-level labels are accessible without per-instance annotations (Shao et al., 2025; Xiao et al., 2026). Existing MIL approaches can be systematically divided into three categories: heuristic instance selection methods, traditional statistical MIL models, and modern deep attention-based pipelines. Early heuristic MIL algorithms rely on handcrafted rules to select representative instances for prediction (Briggs et al., 2012; Settles et al., 2007), which lack generalization on complex real-world data. Classical statistical MIL methods model bag distributions via Gaussian kernels, yet they fail to scale to high-dimensional features (Maron & Ratan, 1998). Recent deep MIL frameworks adopt attention pooling to aggregate instance embeddings into unified bag representations (Cui et al., 2023; Dong et al., 2025; Ilse et al., 2018; Zheng et al., 2024). Although a small number of deep MIL models output separate instance and bag feature branches, they merely maintain two independent prediction streams without cross-granular fusion and iterative mutual optimization between multi-scale representations. Moreover, most existing deep MIL designs ignore two critical real-world challenges: severe intra-bag class imbalance and label noise from ambiguous bag annotations. A series of recent work studies imbalance-aware MIL loss design and noise-robust instance weighting (Shi et al., 2020), but they cannot handle candidate label sets with multiple false positive tags, which restricts their application to MIPL tasks.

2.2. Partial label learning

Partial label learning addresses label-space ambiguity where each sample is assigned a candidate set containing one true label and multiple false positives (Deng et al., 2026; Souriau et al., 2026; Wang et al., 2025). We categorize mainstream PLL algorithms into four branches: progressive true label identification, consistency regularization, weighted loss disambiguation, and non-candidate suppression. Representative progressive PLL methods gradually filter out false labels via model confidence (Lv et al., 2020); consistency-based PLL enforces identical predictions over augmented inputs to refine pseudo-labels (Feng et al., 2020); weighted loss schemes assign lower loss weights to low-confidence candidate labels to reduce noise interference (Wen et al., 2021); non-candidate penalty terms directly suppress probabilities of labels outside candidate sets (Wu et al., 2022). Nevertheless, all existing PLL techniques are designed for single-instance data and cannot be extended to multi-instance bags with intra-bag instance imbalance. Recent studies explore noise-robust and imbalance adaptive PLL

losses (Qiao et al., 2025; Wang et al., 2023), but they do not model the hierarchical bag-instance structure inherent in MIPL, making them incompatible with our task setting.

2.3. Class imbalance

Apart from the above partial-label learning studies, recent advanced algorithms for class imbalance and noise-robust learning are mostly designed for single-sample scenarios and cannot adapt to MIPL's bag-instance hierarchical structure and dual ambiguity. Chen et al. (2026) propose an incremental active learning framework with double uncertainty weighting to handle multi-class imbalance, yet it only processes independent individual samples without multi-instance bags. For imbalanced datasets mixed with outliers, Chen et al. (2025) combine graph embedding and intuitionistic fuzzy broad networks to distinguish clean and noisy data, but this scheme lacks support for partial label candidate sets. Pan et al. (2026) put forward an overlap-driven feature selection metric to mitigate skewed class distribution, which ignores the coexistence of instance ambiguity and label ambiguity unique to MIPL. Unlike these single-sample imbalance learning works, our method targets multi-instance hierarchical data to jointly alleviate intra-bag imbalance and candidate label noise through multi-granular feature and manifold collaborative optimization.

2.4. Multi-instance partial-label learning

As a dual weak supervision paradigm integrating the instance ambiguity of MIL and label ambiguity of PLL, MIPL differs fundamentally from the two single-source weakly supervised paradigms: standard MIL is equipped with definite bag-level labels without label fuzziness, while PLL only handles ambiguous labels for individual samples, yet MIPL bears twofold uncertainty where each bag contains unannotated instances paired with a set of ambiguous candidate labels, making it intractable to be resolved by pure MIL or PLL solutions. We systematically review four representative state-of-the-art MIPL algorithms: MIPLGP (Tang et al., 2024c) adapts Gaussian process modeling for multi-instance bags with candidate labels but merely adopts aggregated bag features and abandons fine-grained instance cues; DEMIPL (Tang et al., 2023) leverages momentum attention embedding for label disambiguation with all feature learning restricted to bag aggregation and no dedicated noise removal module; ELIMIPL (Tang et al., 2024b) enhances disambiguation via conjugate label information but is still confined to single-granularity bag representations without adaptive instance screening; MIPLMA (Tang et al., 2024a) constructs a margin-adjusted loss to relieve label bias. The reviewed representative methods do not explicitly integrate a dynamically purified instance summary and a global bag representation for manifold-based label refinement, and epoch-adaptive instance screening remains less explored in this setting. We thus propose HFC-MIPL, which builds a hybrid representation, integrates focusing and manifold dual disambiguation, and utilizes attention-based adaptive filtering to jointly resolve intra-bag imbalance and candidate label noise.

3. Methodology

3.1. Preliminaries

In multi-instance partial-label learning (MIPL), the training set consists of m multi-instance bags denoted as

$$\mathcal{D} = \{(X_i, S_i)\}_{i=1}^m,$$

where each bag $X_i = \{\mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, \mathbf{x}_{in_i}\}$ contains n_i instances, and $\mathbf{x}_{ij} \in \mathbb{R}^d$ is the j th instance in the i th bag with d -dimensional feature vector. $S_i \subseteq \mathcal{Y}$ is the candidate label set associated with bag X_i , where $\mathcal{Y} = \{1, 2, \dots, K\}$ denotes the label space with K classes. Following the

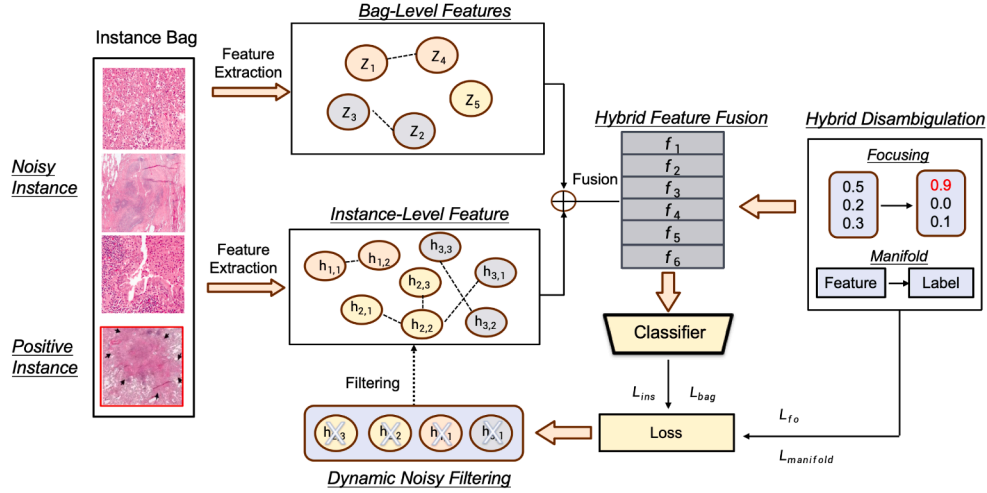


Fig. 1. The framework of HFC-MIPL.

standard MIPL setting (Tang et al., 2023, 2024b), each bag holds exactly one *ground-truth* label $y_i \in S_i$, which is unknown during training. Let $a_i = [a_{i1}, a_{i2}, \dots, a_{in_i}]$ collect the raw sigmoid attention scores, where $a_{ij} \in (0, 1)$. The coefficient used for bag aggregation is $a_{ij} / \sum_{t=1}^{n_i} a_{it}$, which is normalized within each bag and sums to one. Let z_i denote the bag-level representation aggregated from instance features. Let p_i denote the predictive label distribution over bag X_i , and p_{ij} denote the predictive distribution over instance x_{ij} . The goal of MIPL is to learn a prediction model that can recover the true label y_i for each bag X_i by performing disambiguation on the candidate set S_i , while exploiting the discriminative information from instances inside each bag.

The pipeline of the proposed HFC-MIPL is illustrated in Fig. 1, which consists of four core modules: hybrid feature representation, dual disambiguation strategy, unified collaborative training framework, and dynamic noisy instance filtering. Given a multi-instance bag X_i with its candidate label set S_i , we first use a feature extractor to obtain instance-level features and further aggregate them into bag-level representations via attention pooling. We then fuse instance-level and bag-level features to construct hybrid features that preserve both fine-grained details and global semantics. Next, the dual disambiguation strategy, including focusing regularization and manifold regularization, is used to shape the learned distributions and refine pseudo-label information. Finally, the unified framework jointly optimizes the bag and instance branches, while the attention-based threshold determines the purified instance summary used in hybrid-feature construction. By integrating the above modules, HFC-MIPL effectively alleviates dual ambiguity in both instance space and label space and achieves robust prediction for multi-instance partial-label learning.

3.2. Hybrid feature extraction

We construct a hybrid feature representation that retains a purified instance summary together with a global bag representation for subsequent graph construction and collaborative optimization. The specific steps are as follows:

3.2.1. Instance-level feature extraction

A feature extraction network $\psi_1(\cdot)$ is used to extract features from each instance x_{ij} to obtain instance-level features h_{ij} . The formula is as follows:

$$h_{ij} = \psi_1(x_{ij}) \in \mathbb{R}^{d'} \quad (1)$$

where d' is the output dimension of instance-level features. $\psi_1(\cdot)$ adopts a two-layer convolutional neural network (for image datasets) or a fully

connected feedforward neural network (for non-image datasets), referring to the feature extraction strategies of DEMIPL to ensure the effectiveness of feature extraction.

3.2.2. Bag-level feature extraction

Instance-level features are aggregated based on the attention mechanism to obtain bag-level aggregated features z_i . First, the attention weight a_{ij} of each instance x_{ij} is calculated with reference to the gated attention mechanism (Shi et al., 2020). The formula can be expressed as:

$$a_{ij} = \sigma(\mathbf{W} \cdot (\tanh(\mathbf{W}_v \mathbf{h}_{ij} + \mathbf{b}_v) \odot \sigma(\mathbf{W}_u \mathbf{h}_{ij} + \mathbf{b}_u))) \quad (2)$$

where $\mathbf{W} \in \mathbb{R}^{k \times 1}$, $\mathbf{W}_v, \mathbf{W}_u \in \mathbb{R}^{d' \times k}$, and $\mathbf{b}_v, \mathbf{b}_u \in \mathbb{R}^k$ are learnable attention parameters; $\tanh(\cdot)$ is the hyperbolic tangent, $\sigma(x) = \frac{1}{1 + \exp(-x)}$ is the sigmoid function, and $\odot$ denotes element-wise multiplication. Eq. (2) produces the raw score $a_{ij} \in (0, 1)$, which is used by the threshold rule.

Subsequently, the instance-level features are aggregated through attention-weighted summation to obtain the bag-level aggregated feature z_i , which can be expressed as:

$$z_i = \sum_{j=1}^{n_i} \left(\frac{a_{ij}}{\sum_{t=1}^{n_i} a_{it}} \right) h_{ij}, \quad z_i \in \mathbb{R}^{d'} \quad (3)$$

The coefficient $a_{ij} / \sum_{t=1}^{n_i} a_{it}$ in Eq. (3) is normalized within each bag and is used only for bag aggregation; it is distinct from the raw score used for thresholding.

3.3. Hybrid feature representation and losses

The hybrid representation combines a purified instance summary with a global bag embedding so that both feature sources are available for graph-based label refinement. We first compute the bag feature z_i as defined in Eqs. (2) and (3). A fully connected transformation network $\psi_2(\cdot)$ maps z_i into a unified dimension:

$$\tilde{z}_i = \psi_2(z_i) \in \mathbb{R}^{d'} \quad (4)$$

During hybrid-feature construction, the current thresholded set $\mathcal{K}_i$ is used to calculate the purified instance mean $\bar{h}_i$:

$$\bar{h}_i = \frac{1}{N_i} \sum_{j \in \mathcal{K}_i} h_{ij} \quad (5)$$

where N_i is the number of instances in $\mathcal{K}_i$, and $\mathcal{K}_i$ contains the instances whose raw attention scores meet the current threshold τ_i . Instances outside $\mathcal{K}_i$ do not enter this purified mean at the current stage, whereas the bag representation z_i remains a normalized attention-weighted aggregation over the complete bag.

We use dimension-aligned concatenation as a simple, parameter-free design choice that preserves both feature sources before graph construction:

$$f_i = [\tilde{z}_i; \tilde{h}_i] \in \mathbb{R}^{2d'} \quad (6)$$

The resulting hybrid feature is used to construct the similarity graph for manifold-based label disambiguation, while the bag and instance prediction branches retain their respective inputs for joint optimization.

3.3.1. Bag-level cross-entropy loss

The bag prediction branch takes attention-aggregated feature z_i as input and outputs label distribution P_i . We apply masked cross-entropy only over candidate label set S_i to supervise global bag classification:

$$L_{\text{bag}} = -\frac{1}{m} \sum_{i=1}^m \sum_{c \in S_i} \log(P_{i,c}) \quad (7)$$

where $P_{i,c}$ represents the predicted probability of assigning label c to bag X_i . The candidate-set term provides candidate-constrained weak supervision: it excludes non-candidate labels from the loss sum but does not by itself identify the unique true label in S_i . Label differentiation is handled jointly with manifold-refined pseudo-label information, focusing regularization, and the attention-based instance branch.

3.3.2. Instance-weighted cross-entropy loss

Because instance-level ground-truth labels are unavailable, all instances in a bag share the bag's candidate set as weak supervision, and a_{ij} modulates their relative contributions. The weighted instance loss is formulated as:

$$L_{\text{ins}} = -\frac{1}{m} \sum_{i=1}^m \sum_{j=1}^{n_i} a_{ij} \sum_{c \in S_i} \log(P_{ij,c}) \quad (8)$$

Here $P_{ij,c}$ denotes the predicted label probability for instance x_{ij} . This term is an auxiliary weak-supervision objective; the attention coefficient modulates relative contributions and is not interpreted as ground-truth supervision or as a guarantee that only positive instances contribute.

3.4. Dual disambiguation strategy

According to the characteristics of MIPL tasks—dual ambiguity, we design a dual disambiguation strategy to address these two types of ambiguity simultaneously, which further enhances the reliability of label disambiguation and feature selection. The strategy consists of two complementary components: focusing regularization and manifold regularization, which respectively focus on weight optimization and label similarity propagation.

3.4.1. Focusing regularization

The focusing terms are introduced as auxiliary distribution-shaping regularizers during joint optimization; they modulate the attention and candidate-label distributions together with the other loss components.

Inspired by the sparse regularization term in related work (Wen et al., 2021), we define the following attention-distribution regularizer:

$$L_{\text{sp-feat}} = \frac{1}{m} \sum_{i=1}^m \sum_{j=1}^{n_i} a_{ij}^T \log(a_{ij}^T) \quad (9)$$

where a_{ij} denotes the attention score of the j th instance in the i th bag, and $T > 1$ is the focusing factor. This term shapes the attention distribution jointly with the remaining objectives; it is not claimed to identify key instances independently.

For candidate-label distribution shaping, we use the following L1-based term:

$$L_{\text{sp-label}} = \frac{1}{m} \sum_{i=1}^m \|P_i \odot S_i\|_1 \quad (10)$$

where S_i denotes the candidate-label indicator, $\odot$ denotes element-wise multiplication, and $\|\cdot\|_1$ represents the L1 norm. This term is optimized

jointly with the other loss components and is not claimed to recover the true label independently. The total focusing regularization loss is the weighted sum of the two terms:

$$L_{fo} = L_{\text{sp-feat}} + \beta L_{\text{sp-label}} \quad (11)$$

where β is a positive hyperparameter, which is used to balance the regularization strengths of attention weight focusing and label probability focusing in the joint training process.

3.4.2. Manifold regularization

Based on the manifold learning assumption, instances with similar hybrid features tend to yield consistent label predictions (Wang et al., 2023). To propagate feature similarity information from the hybrid feature space to the label space for robust label disambiguation in MIPL, we construct a unified similarity graph based on fused hybrid features.

Specifically, we compute the fused hybrid features similarity matrix S using cosine similarity with K -nearest neighbor sparsification:

$$S_{pq} = \begin{cases} \max(f_p^T f_q, 0)^\gamma, & q \in \mathcal{K}_K(f_p), \\ 0, & \text{otherwise,} \end{cases} \quad (12)$$

where f_p and f_q denote fused hybrid features (consistent with the hybrid feature symbol f_i defined above), $\mathcal{K}_K(\cdot)$ represents the K -nearest neighbor set, and $\gamma > 0$ controls the similarity weight. The graph is symmetrized as $H = S + S^T$, and its normalized graph Laplacian is calculated as:

$$\Delta = E - \Lambda^{-\frac{1}{2}} H \Lambda^{-\frac{1}{2}},$$

where Λ is the degree matrix of H .

We introduce graph Laplacian regularization to propagate the structural similarity of hybrid features to the label prediction space, ensuring consistent label assignments for similar features:

$$\mathcal{R}(\tilde{Y}) = \text{tr}(\tilde{Y}^T \Delta \tilde{Y}), \quad (13)$$

where $\tilde{Y}$ denotes the label prediction matrix.

By integrating a label consistency term that aligns predictions with pseudo-label estimates, the overall objective function is formulated as:

$$\mathcal{Z}(\tilde{Y}, Y^*) = \mu \|\tilde{Y} - Y^*\|_F^2 + \kappa \mathcal{R}(\tilde{Y}), \quad (14)$$

where Y^* is the current pseudo-label matrix, and μ, κ are hyperparameters that balance consistency and smoothness.

To obtain stable and reliable pseudo-labels, we solve the following optimization problem:

$$\min_Q \mathcal{L}_{\text{LP}}(Q) = \frac{1}{2} \|Q - \tilde{Y}\|_F^2 + \mathcal{Z}(Q, Y^*), \quad (15)$$

and the corresponding gradient is derived as:

$$\frac{\partial \mathcal{L}_{\text{LP}}}{\partial Q} = (1 + 2\mu)Q + 2\kappa \Delta Q - (\tilde{Y} + 2\mu Y^*). \quad (16)$$

Finally, the pseudo-label matrix Q is updated via lightweight gradient descent:

$$Q = Q - \eta_l \cdot \frac{\partial \mathcal{L}_{\text{LP}}}{\partial Q}, \quad (17)$$

where η_l denotes the learning rate. This framework effectively propagates structural information from the hybrid feature manifold to the label space, providing reliable supervision for dual disambiguation in MIPL tasks.

After obtaining the refined pseudo-labels Y^* through graph-based label propagation, we optimize the model to fit these reliable supervisions using the Probabilistic Mean Squared Error (PMSE) loss:

$$L_{\text{manifold}}(\tilde{Y}, Y^*) = \frac{1}{m} \sum_{i=1}^m \|\sigma(\tilde{Y}_i) - Y_i^*\|_2^2, \quad (18)$$

where $\sigma(\cdot)$ denotes the sigmoid function for probability normalization (consistent with the definition above), $\tilde{Y}_i$ is the model's predicted label distribution, and Y_i^* represents the refined pseudo-label after manifold propagation. This loss ensures the model predictions are consistent with the propagated pseudo-labels, further boosting the accuracy of label disambiguation.

3.5. Unified training framework and dynamic noisy instance filtering

To achieve collaborative optimization between the disambiguation strategy and model learning, we propose a unified training framework integrated with dynamic noisy instance filtering.

3.5.1. Loss function

To jointly exploit bag-level global semantics and instance-level fine-grained cues for label disambiguation, we design a unified composite loss function consisting of four complementary components. We compute two sets of predictive distributions, respectively, from the bag-aggregated feature z_i and raw instance feature $h_{i,j}$, which serve as the inputs for bag loss and instance loss calculation separately. The overall fused objective $\mathcal{L}_{\text{total}}$ integrates bag classification loss $\mathcal{L}_{\text{bag}}$, attention-weighted instance classification loss $\mathcal{L}_{\text{ins}}$, focusing regularization loss $\mathcal{L}_{\text{fo}}$, and manifold consistency loss $\mathcal{L}_{\text{manifold}}$. The unified total loss is formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{bag}} + \lambda_1 \mathcal{L}_{\text{ins}} + \lambda_2 \mathcal{L}_{\text{fo}} + \lambda_3 \mathcal{L}_{\text{manifold}} \quad (19)$$

where $\lambda_1, \lambda_2 > 0, \lambda_3 > 0$ are trade-off hyperparameters to balance the contribution of each loss term. Concretely, $\mathcal{L}_{\text{bag}}$ is the masked cross-entropy loss defined in Eq. (7) for bag supervision; $\mathcal{L}_{\text{ins}}$ denotes the weighted instance cross-entropy loss defined in Eq. (8), which is only activated after half of the total training epochs according to our implementation. $\mathcal{L}_{\text{fo}}$ represents the combined focusing penalty given in Eq. (11), and $\mathcal{L}_{\text{manifold}}$ is the manifold alignment loss presented in Eq. (18).

3.5.2. Model updating

Backpropagation jointly updates the prediction and attention parameters under the composite objective, while manifold-refined pseudo-label information contributes through the manifold loss.

The updated raw attention scores are then used with the current threshold to recompute the retained set for purified-feature construction; this process is not interpreted as permanent removal from every learning branch.

3.5.3. Dynamic noisy instance filtering

To mitigate the negative impact of low-quality instances, we design a dynamic filtering mechanism driven by attention weights. Specifically, we employ an adaptive attention threshold τ_t that dynamically evolves with training epoch t :

$$\tau_t = \min \left(\tau_0 + \Delta \cdot \left\lfloor \frac{t}{T} \right\rfloor, \tau_{\text{max}} \right), \quad (19)$$

where τ_0 is the initial threshold, Δ controls the growth step, T denotes the interval of threshold update, and τ_{max} is the upper bound. During hybrid-feature construction, the current thresholded set $\mathcal{K}_i$ is used to compute the purified mean $\bar{h}_i$. Instances outside $\mathcal{K}_i$ do not enter this purified mean at the current stage, whereas the bag representation z_i remains a normalized attention-weighted aggregation over the complete bag. The retained set is recomputed from the current scores and threshold, so the screening decision is branch-specific rather than a permanent deletion from all subsequent computations. This schedule reduces the risk of premature exclusion but does not guarantee retention of every rare positive instance under extreme intra-bag imbalance; controlled evaluation across positive-instance ratios is left for future work.

Algorithm (1) summarizes the complete training flow of the HFC-MIPL method proposed in this paper. At each iteration, the current thresholded set is formed before the purified instance mean and hybrid feature are constructed, after which the prediction losses and graph-based refinement are updated.

4. Experiments

This section comprehensively evaluates the performance of the proposed method on benchmark datasets.

Algorithm 1 Training procedure of HFC-MIPL.

Require: Training dataset $\mathcal{D} = \{(\mathbf{X}_i, \mathbf{S}_i)\}_{i=1}^m$, feature extractor ϕ , attention parameters $\{\mathbf{W}, \mathbf{W}_v, \mathbf{W}_u, \mathbf{b}_v, \mathbf{b}_u\}$, loss weights $\{\lambda_1, \lambda_2, \lambda_3\}$, initial threshold $\tau_0 = 0.2$, learning rate γ , total epochs T , graph construction interval Δt

```

1: Initialize all model parameters  $\Theta$  uniformly
2: for  $t = 1$  to  $T$  do
3:   for each bag  $\mathbf{X}_i \in \mathcal{B}$  do
4:     Extract instance features:  $\mathbf{h}_{i,j} = \phi(\mathbf{x}_{i,j})$  (Eq. 1)
5:     Calculate instance attention:  $a_{i,j}$  (Eq. 2)
6:     Aggregate to get bag-level feature:  $\tilde{z}_i$  (Eq. 4)
7:     Form the current retained set:  $\mathcal{K}_i = \{j \mid a_{i,j} \geq \tau_t\}$  (Eq. 19)
8:     Compute the purified instance mean  $\bar{h}_i$  (Eq. 5)
9:     Construct the hybrid feature  $\mathbf{f}_i$  (Eq. 6)
10:    Compute total loss (Eq. 19):
         $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{bag}} + \lambda_1 \mathcal{L}_{\text{ins}} + \lambda_2 \mathcal{L}_{\text{fo}} + \lambda_3 \mathcal{L}_{\text{manifold}}$ 
11:    Update parameters:  $\Theta \leftarrow \Theta - \gamma \nabla_{\Theta} \mathcal{L}_{\text{total}}$ 
12:  end for
13:  if  $t \geq \Delta t$  then
14:    Construct similarity graph  $\mathbf{H}$  from hybrid features  $\{\mathbf{f}_i\}$  (Eq. 12)
15:    Compute graph Laplacian  $\Delta$  and manifold regularization
         $\mathcal{L}_{\text{manifold}}$  (Eq. 18)
16:    Perform label propagation to get disambiguated pseudo-labels
         $\mathbf{Q}$  (Eq. 17)
17:  end if
18: end for
Ensure: Trained model  $\Theta$  and bag-level predictions  $\hat{\mathbf{P}}$ 

```

4.1. Experimental setup

Datasets. We use four benchmark datasets—MNIST-MIPL, FMNIST-MIPL, Birdsong-MIPL, and SIVAL-MIPL—and one real-world medical dataset, CRC-MIPL. These datasets cover image, audio, and medical domains (Briggs et al., 2012; Kaye et al., 2020; LeCun et al., 2002; Settles et al., 2007), and each follows the multi-instance partial-label setting where each bag contains exactly one true label and several false positive candidates. We use multi-instance features generated by four bag generators: Row (Wei & Zhou, 2016), Single Blob with Neighbors (SBN) (Maron & Ratan, 1998), KMeans Segmentation (KMeansSeg) (Yildizer et al., 2012), and Scale-Invariant Feature Transform (SIFT) (Lowe, 2004). The key statistics of the datasets are summarized in Table 1. To evaluate the performance of HFC-MIPL under different levels of label ambiguity, we vary the number of false positive labels $r \in \{1, 2, 3\}$ in all benchmark datasets. For the real-world medical dataset, we use four different feature extraction methods to evaluate HFC-MIPL under feature variations.

4.1.1. Comparative methods

We compare HFC-MIPL with eight representative MIPL and PLL baselines, including four MIPL-specific algorithms and four partial-label learning algorithms: 1) **Existing MIPL algorithms:** MIPLGP (Tang et al., 2024c), DEMIPL (Tang et al., 2023), MIPLMA (Tang et al., 2024a) and ELIMIPL (Tang et al., 2024b); 2) **state-of-the-art PLL algorithms:** PRO-DEN (Lv et al., 2020), RC (Feng et al., 2020), LWS (Wen et al., 2021), and PL-AGGD (Wang et al., 2022). All compared methods were evaluated under the common data construction and evaluation protocol used in this study. We followed the settings reported in the corresponding papers where applicable.

4.1.2. Parameter settings

HFC-MIPL runs on a workstation with 8 NVIDIA RTX 3090 GPUs. The hyperparameters of our method are set based on Tang et al. (2023, 2024b) and fine-tuned via grid search. The optimizer is SGD with momentum 0.9 and weight decay 10^{-4} . The learning rate is selected from

Table 1
Statistics of the benchmark and real-world MIPL datasets.

Dataset	#Bags	#Instances	Avg.#Inst/Bag	#Dim	#Class	Avg.#Cls	Domain
MNIST-MIPL	500	20,664	41.33	784	5	2–4	Image
FMNIST-MIPL	500	20,810	41.62	784	5	2–4	Image
Birdsong-MIPL	1300	48,425	37.25	38	13	2–4	Audio
SIVAL-MIPL	1500	47,414	31.61	30	25	2–4	Image
CRC-MIPL	7000	56,000-175,000	6–25	9–128	7	2.08	Medical

Table 2
Classification accuracies (mean $\pm$ std) of HFC-MIPL and comparative algorithms on MNIST-MIPL and FMNIST-MIPL with varying numbers of false positive candidate labels ($r = \{1, 2, 3\}$).

Method	MNIST-MIPL			FMNIST-MIPL		
	$r = 1$	$r = 2$	$r = 3$	$r = 1$	$r = 2$	$r = 3$
HFC-MIPL (Ours)	0.991 $\pm$ 0.007	0.980 $\pm$ 0.014	0.786 $\pm$ 0.068	0.916 $\pm$ 0.018	0.813 $\pm$ 0.023	0.611 $\pm$ 0.053
DEMIP	0.976 $\pm$ 0.008	0.952 $\pm$ 0.019	0.677 $\pm$ 0.125	0.856 $\pm$ 0.016	0.791 $\pm$ 0.034	0.596 $\pm$ 0.127
MIPLGP	0.949 $\pm$ 0.016	0.817 $\pm$ 0.030	0.621 $\pm$ 0.064	0.847 $\pm$ 0.030	0.791 $\pm$ 0.027	0.670 $\pm$ 0.052
MIPLMA	0.983 $\pm$ 0.009	0.975 $\pm$ 0.016	0.762 $\pm$ 0.105	0.884 $\pm$ 0.024	0.698 $\pm$ 0.197	0.298 $\pm$ 0.099
ELIMIPL	0.983 $\pm$ 0.231	0.946 $\pm$ 0.038	0.724 $\pm$ 0.037	0.897 $\pm$ 0.016	0.731 $\pm$ 0.006	0.619 $\pm$ 0.092
PRODEN + Mean	0.605 $\pm$ 0.023	0.481 $\pm$ 0.036	0.283 $\pm$ 0.028	0.697 $\pm$ 0.042	0.573 $\pm$ 0.026	0.345 $\pm$ 0.027
RC + Mean	0.658 $\pm$ 0.031	0.598 $\pm$ 0.033	0.392 $\pm$ 0.033	0.753 $\pm$ 0.042	0.649 $\pm$ 0.028	0.401 $\pm$ 0.063
LWS + Mean	0.463 $\pm$ 0.048	0.209 $\pm$ 0.028	0.205 $\pm$ 0.013	0.726 $\pm$ 0.031	0.720 $\pm$ 0.025	0.579 $\pm$ 0.041
PL-AGGD + Mean	0.671 $\pm$ 0.027	0.595 $\pm$ 0.036	0.380 $\pm$ 0.032	0.743 $\pm$ 0.026	0.677 $\pm$ 0.028	0.474 $\pm$ 0.057
PRODEN + MaxMin	0.508 $\pm$ 0.024	0.400 $\pm$ 0.037	0.345 $\pm$ 0.048	0.424 $\pm$ 0.045	0.377 $\pm$ 0.040	0.309 $\pm$ 0.058
RC + MaxMin	0.519 $\pm$ 0.028	0.469 $\pm$ 0.035	0.380 $\pm$ 0.048	0.731 $\pm$ 0.027	0.666 $\pm$ 0.027	0.524 $\pm$ 0.034
LWS + MaxMin	0.242 $\pm$ 0.042	0.239 $\pm$ 0.048	0.218 $\pm$ 0.017	0.435 $\pm$ 0.049	0.406 $\pm$ 0.040	0.318 $\pm$ 0.064
PL-AGGD + MaxMin	0.527 $\pm$ 0.035	0.439 $\pm$ 0.020	0.321 $\pm$ 0.043	0.391 $\pm$ 0.040	0.371 $\pm$ 0.037	0.327 $\pm$ 0.028

$\{0.01, 0.05, 0.001\}$. The maximum training epoch is set to 300. The manifold parameter $k = 10$, and the initial noisy elimination threshold $\tau_0 = 0.2$ with a dynamic increase strategy. To ensure the reliability of the results, the classification accuracy (mean $\pm$ standard deviation) of 10 independent train/test splits (7:3) is reported. The results are presented in two tables, with the highest accuracy in each column marked in bold to facilitate comparison.

4.2. Results on the benchmark datasets

To evaluate HFC-MIPL, we conduct experiments on MNIST-MIPL, FMNIST-MIPL, Birdsong-MIPL, and SIVAL-MIPL with $r \in \{1, 2, 3\}$. The comparison uses representative MIPL algorithms (DEMIP, MIPLGP, MIPLMA, and ELIMIPL) and PLL baselines with Mean and MaxMin decoding; the complete results are reported in Tables 2 and 3.

HFC-MIPL obtains the highest reported accuracy for all three MNIST-MIPL settings. On FMNIST-MIPL, it ranks first at $r = 1$ (0.916 ± 0.018) and $r = 2$ (0.813 ± 0.023), while MIPLGP is higher at $r = 3$. On SIVAL-MIPL, HFC-MIPL ranks first at $r = 1$ and $r = 2$, whereas ELIMIPL is higher at $r = 3$. On Birdsong-MIPL, HFC-MIPL ranks first at $r = 1$ and $r = 3$ and second to MIPLGP at $r = 2$.

Overall, HFC-MIPL achieves the best or second-best result in most evaluated settings, but it is not uniformly ranked first.

4.3. Results on real-world dataset

To further validate the practical applicability and robustness of our proposed HFC-MIPL method in real-world clinical scenarios, we conduct comprehensive experiments on the CRC-MIPL medical dataset. All comparative methods are optimized under the identical experimental protocol to ensure a fair and consistent comparison. As summarized in Table 4, HFC-MIPL achieves competitive performance across all four feature extraction paradigms, with clear advantages in most scenarios, fully demonstrating its practical value in real-world medical diagnosis tasks.

Overall, HFC-MIPL outperforms the majority of MIPL and PLL baselines across all feature types. Specifically, on CRC-MIPL (Row), HFC-MIPL (0.416 ± 0.007) is slightly lower than MIPLGP, but still outperforms DEMIP, ELIMIPL and MIPLMA by noticeable margins. On CRC-MIPL (SBN), HFC-MIPL (0.491 ± 0.011) achieves a significant lead over all MIPL baselines including MIPLGP. For the two more challenging feature settings, HFC-MIPL secures the optimal performance: on CRC-MIPL (KMeansSeg), HFC-MIPL (0.493 ± 0.017) ranks first among all MIPL methods, and on CRC-MIPL (SIFT), HFC-MIPL (0.542 ± 0.011) outperforms all competitors by a clear gap.

Among the MIPL baselines, DEMIP delivers stable performance across all feature types, but is consistently surpassed by HFC-MIPL on SBN and SIFT features. The newly added MIPLMA and ELIMIPL show limited performance on most feature settings, with only ELIMIPL achieving competitive results on the SIFT feature, which further highlights the superiority of our hybrid feature design. For PLL baselines, PL-AGGD + MaxMin achieves the highest accuracy on Row, SBN and KMeansSeg features, while HFC-MIPL maintains highly competitive performance and outperforms all other PLL variants. The remaining PLL methods show degraded performance on most feature settings, which further confirms that MIPL-specific modeling is more suitable for this real-world medical multi-instance partial-label task.

In summary, the experimental results on the CRC-MIPL dataset demonstrate that HFC-MIPL exhibits strong adaptability to diverse feature extraction methods, maintains state-of-the-art performance compared to existing MIPL and PLL baselines, and has promising practical application potential in real-world healthcare scenarios.

4.4. Ablation study

To verify the effectiveness of each core component in the proposed HFC-MIPL method, we conduct a comprehensive ablation study on the FMNIST-MIPL dataset, with label ambiguity controlled by $r \in \{1, 2, 3\}$. We adopt a component-removal paradigm to evaluate individual component contributions, constructing three ablated variants: (1) HFC-MIPL w/o Dynamic Noisy Instance Filtering (removing the noisy instance fil-

Table 3

Classification accuracies (mean $\pm$ std) of HFC-MIPL and comparative algorithms on Birdsong-MIPL and SIVAL-MIPL with varying numbers of false positive candidate labels ($r = \{1, 2, 3\}$).

Method	Birdsong-MIPL			SIVAL-MIPL		
	$r = 1$	$r = 2$	$r = 3$	$r = 1$	$r = 2$	$r = 3$
HFC-MIPL (Ours)	0.718 $\pm$ 0.019	0.660 $\pm$ 0.014	0.637 $\pm$ 0.018	0.601 $\pm$ 0.034	0.510 $\pm$ 0.033	0.453 $\pm$ 0.026
DEMIPL	0.696 $\pm$ 0.029	0.634 $\pm$ 0.019	0.622 $\pm$ 0.025	0.551 $\pm$ 0.017	0.347 $\pm$ 0.193	0.346 $\pm$ 0.076
MIPLGP	0.716 $\pm$ 0.026	0.672 $\pm$ 0.015	0.625 $\pm$ 0.015	0.579 $\pm$ 0.037	0.489 $\pm$ 0.031	0.422 $\pm$ 0.027
MIPLMA	0.618 $\pm$ 0.023	0.508 $\pm$ 0.040	0.470 $\pm$ 0.039	0.427 $\pm$ 0.030	0.292 $\pm$ 0.035	0.230 $\pm$ 0.024
ELIMIPL	0.711 $\pm$ 0.015	0.698 $\pm$ 0.013	0.620 $\pm$ 0.018	0.599 $\pm$ 0.024	0.509 $\pm$ 0.021	0.503 $\pm$ 0.022
PRODEN + Mean	0.296 $\pm$ 0.014	0.272 $\pm$ 0.019	0.211 $\pm$ 0.013	0.219 $\pm$ 0.014	0.184 $\pm$ 0.014	0.166 $\pm$ 0.017
RC + Mean	0.362 $\pm$ 0.015	0.335 $\pm$ 0.011	0.298 $\pm$ 0.009	0.279 $\pm$ 0.011	0.258 $\pm$ 0.017	0.237 $\pm$ 0.020
LWS + Mean	0.265 $\pm$ 0.010	0.254 $\pm$ 0.010	0.237 $\pm$ 0.005	0.240 $\pm$ 0.014	0.223 $\pm$ 0.008	0.194 $\pm$ 0.026
PL-AGGD + Mean	0.353 $\pm$ 0.019	0.314 $\pm$ 0.018	0.296 $\pm$ 0.015	0.355 $\pm$ 0.015	0.315 $\pm$ 0.019	0.286 $\pm$ 0.018
PRODEN + MaxMin	0.387 $\pm$ 0.014	0.357 $\pm$ 0.012	0.336 $\pm$ 0.012	0.316 $\pm$ 0.019	0.287 $\pm$ 0.024	0.250 $\pm$ 0.018
RC + MaxMin	0.390 $\pm$ 0.014	0.371 $\pm$ 0.013	0.363 $\pm$ 0.010	0.306 $\pm$ 0.023	0.288 $\pm$ 0.021	0.267 $\pm$ 0.020
LWS + MaxMin	0.225 $\pm$ 0.038	0.207 $\pm$ 0.034	0.216 $\pm$ 0.029	0.289 $\pm$ 0.017	0.271 $\pm$ 0.014	0.244 $\pm$ 0.023
PL-AGGD + MaxMin	0.383 $\pm$ 0.014	0.372 $\pm$ 0.020	0.344 $\pm$ 0.011	0.397 $\pm$ 0.028	0.360 $\pm$ 0.029	0.328 $\pm$ 0.023

Table 4

Classification accuracies (mean $\pm$ std) of HFC-MIPL and comparative algorithms on CRC-MIPL with different feature extraction methods.

Method	CRC-MIPL (Row)	CRC-MIPL (SBN)	CRC-MIPL (KMeansSeg)	CRC-MIPL (SIFT)
HFC-MIPL (Ours)	0.416 $\pm$ 0.007	0.491 $\pm$ 0.011	0.493 $\pm$ 0.017	0.542 $\pm$ 0.011
DEMIPL	0.407 $\pm$ 0.010	0.473 $\pm$ 0.009	0.490 $\pm$ 0.017	0.530 $\pm$ 0.014
MIPLGP	0.432 $\pm$ 0.005	0.335 $\pm$ 0.006	0.329 $\pm$ 0.012	0.419 $\pm$ 0.027
MIPLMA	0.273 $\pm$ 0.008	0.149 $\pm$ 0.012	0.153 $\pm$ 0.016	0.142 $\pm$ 0.010
ELIMIPL	0.372 $\pm$ 0.009	0.137 $\pm$ 0.024	0.326 $\pm$ 0.007	0.503 $\pm$ 0.022
PRODEN + Mean	0.365 $\pm$ 0.009	0.392 $\pm$ 0.008	0.233 $\pm$ 0.018	0.334 $\pm$ 0.029
RC + Mean	0.214 $\pm$ 0.011	0.242 $\pm$ 0.012	0.226 $\pm$ 0.009	0.209 $\pm$ 0.007
LWS + Mean	0.291 $\pm$ 0.010	0.310 $\pm$ 0.006	0.237 $\pm$ 0.008	0.270 $\pm$ 0.007
PL-AGGD + Mean	0.412 $\pm$ 0.008	0.480 $\pm$ 0.005	0.358 $\pm$ 0.008	0.363 $\pm$ 0.012
PRODEN + MaxMin	0.401 $\pm$ 0.007	0.447 $\pm$ 0.011	0.265 $\pm$ 0.027	0.291 $\pm$ 0.011
RC + MaxMin	0.227 $\pm$ 0.012	0.338 $\pm$ 0.010	0.208 $\pm$ 0.007	0.246 $\pm$ 0.008
LWS + MaxMin	0.299 $\pm$ 0.008	0.382 $\pm$ 0.009	0.247 $\pm$ 0.005	0.230 $\pm$ 0.007
PL-AGGD + MaxMin	0.460 $\pm$ 0.008	0.524 $\pm$ 0.008	0.434 $\pm$ 0.009	0.285 $\pm$ 0.009

Table 5

Classification accuracies (mean $\pm$ std) of HFC-MIPL and its ablated variants on FMNIST-MIPL with varying numbers of false positive candidate labels ($r = \{1, 2, 3\}$).

Model	$r = 1$	$r = 2$	$r = 3$
HFC-MIPL (Ours) [ALL]	0.916 $\pm$ 0.018	0.813 $\pm$ 0.023	0.611 $\pm$ 0.053
w/o Dynamic Noisy Instance Filtering	0.898 $\pm$ 0.016	0.810 $\pm$ 0.022	0.605 $\pm$ 0.052
w/o Dual Disambiguation Strategy	0.897 $\pm$ 0.013	0.805 $\pm$ 0.029	0.607 $\pm$ 0.061
w/o Hybrid Feature Fusion + Dynamic Noisy Instance Filtering	0.819 $\pm$ 0.015	0.748 $\pm$ 0.004	0.516 $\pm$ 0.045

tering mechanism); (2) HFC-MIPL w/o Dual Disambiguation Strategy (removing the label ambiguity resolution strategy); (3) HFC-MIPL w/o Hybrid Feature Fusion + Dynamic Noisy Instance Filtering (removing both hybrid feature fusion and noisy instance filtering). Experimental results are reported in Table 5, with the highest accuracy for each r marked in bold.

As shown in Table 5, HFC-MIPL (Ours) [ALL] achieves the highest accuracy across all r values (**0.916 $\pm$ 0.018** for $r = 1$, **0.813 $\pm$ 0.023** for $r = 2$, **0.611 $\pm$ 0.053** for $r = 3$). The component-removal results support the contribution of the evaluated individual components and component combinations under the reported setting.

Notably, removing the dual disambiguation strategy causes an accuracy drop of 0.018 for $r = 1$, 0.003 for $r = 2$, and 0.006 for $r = 3$. Removing the dynamic noisy instance filtering mechanism results in absolute accuracy drops of 0.003–0.008 under the reported settings.

Removing hybrid feature fusion together with noisy instance filtering produces larger absolute drops (0.047–0.097) than removing filtering alone, which supports only their joint contribution and does not isolate

Table 6

Training time per epoch and memory usage of HFC-MIPL and comparative MIPL methods.

Method	Epoch Time (s)	Memory Usage (mb)
HFC-MIPL (Ours)	0.58	422
DEMIPL	0.41	372
MIPLGP	1.03	987

the effect of concatenation. As r increases, all models degrade, but HFC-MIPL maintains an advantage under the reported settings.

4.5. Parameter sensitivity analysis

To verify the robustness of HFC-MIPL to hyperparameters, we conducted sensitivity experiments on FMNIST-MIPL ($r = 1$) with respect to three key parameters λ_1 , λ_2 , and λ_3 . We vary each parameter within a reasonable range while fixing others, and the classification accuracy is reported in Fig. 2.

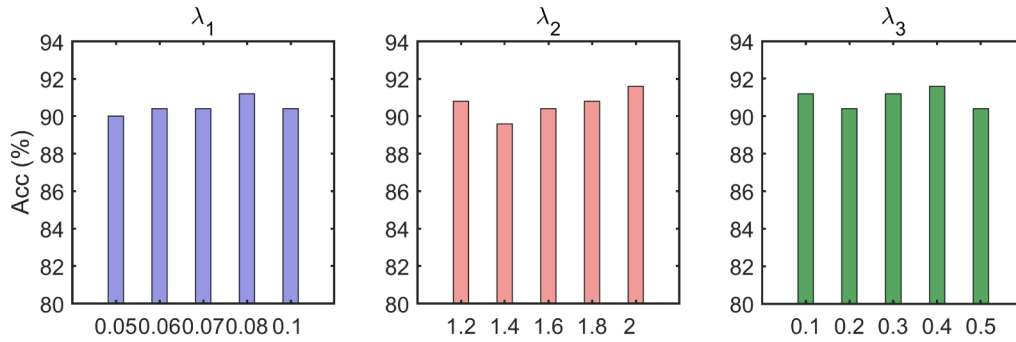


Fig. 2. Parameter sensitivity analysis on FMNIST-MIPL ($r = 1$).

As shown in the results, HFC-MIPL maintains stable and competitive performance across different parameter settings, exhibiting strong robustness. The model achieves optimal performance when $\lambda_1 = 0.08$, $\lambda_2 = 2.0$, and $\lambda_3 = 0.4$, respectively. Performance declines gently when parameters deviate from the optimal values, indicating that HFC-MIPL is not highly sensitive to hyperparameter choices, which facilitates practical deployment.

4.6. Time complexity analysis

To evaluate the computational efficiency of the proposed HFC-MIPL method, we conduct comparative experiments on its training efficiency and memory consumption, with DEMIPL and MIPLGP as the main comparison baselines. All experiments are performed in the same hardware environment (NVIDIA RTX 3090 GPU) to ensure fair comparison. We report the average training time per epoch and memory usage over 10 independent runs, with the results summarized in Table 6.

As illustrated in Table 6, HFC-MIPL achieves a favorable balance between model performance and computational efficiency, outperforming the state-of-the-art MIPL method MIPLGP while maintaining competitive efficiency compared to DEMIPL. Specifically, in terms of training time per epoch, HFC-MIPL consumes 0.58s per epoch, which is slightly higher than DEMIPL (0.41s) but significantly lower than MIPLGP (1.03s), reducing the training overhead by 43.7% compared to MIPLGP. This slight increase in training time compared to DEMIPL is attributed to the additional computational cost introduced by HFC-MIPL's hybrid feature fusion mechanism and dynamic noisy instance elimination, which are critical for its superior performance. In terms of memory usage, HFC-MIPL requires 422mb of memory, which is marginally higher than DEMIPL (372mb) but only 42.7% of MIPLGP's memory consumption (987mb). This efficient memory usage is due to the optimized network structure of HFC-MIPL, which reduces redundant feature storage and instance processing.

5. Conclusion

In this work, we propose a hybrid feature-based collaborative disambiguation method for multi-instance partial-label learning. Our method integrates instance-level and bag-level representations and employs focusing and manifold regularizations to refine learned distributions. Experiments on four benchmark datasets and one real-world medical dataset show that HFC-MIPL achieves competitive performance against the compared MIPL and PLL baselines, while maintaining moderate computational cost. Future work will explore extending our method to more complex real-world MIPL tasks and applying it to broader healthcare and bioinformatics applications.

CRediT authorship contribution statement

Zhen Zhu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project admin-

istration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Yining Sun:** Software, Data curation; **Yuanbo Wang:** Software; **Bingxin Zhou:** Software; **Yiwen Hu:** Software; **YuXing Zhang:** Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Briggs, F., Fern, X. Z., & Raich, R. (2012). Rank-loss support instance machines for MIML instance annotation. In *Proceedings of the 18th ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 534–542).
- Burns, C., Izmailov, P., Kirchner, J. H., Baker, B., Gao, L., Aschenbrenner, L., Chen, Y., Ecoffet, A., Joglekar, M., Leike, J. et al. (2025). Weak-to-strong generalization: Eliciting strong capabilities with weak supervision. In *Forty-first international conference on machine learning*.
- Chen, W., Yang, K., Yu, Z., Nie, F., & Chen, C. L. P. (2025). Adaptive broad network with graph-fuzzy embedding for imbalanced noise data. *IEEE Transactions on Fuzzy Systems*, 33(6), 1949–1962.
- Chen, W., Yu, Z., Yang, K., Fan, Z., & Chen, C. L. P. (2026). Adaptive weighted double uncertainty incrementally active learning for multi-class imbalanced data. *IEEE Transactions on Knowledge & Data Engineering*, 38(02), 827–841.
- Cui, Y., Liu, Z., Liu, X., Liu, X., Wang, C., Kuo, T.-W., Xue, C. J., & Chan, A. B. (2023). Bayes-MIL: A new probabilistic perspective on attention-based multiple instance learning for whole slide images. In *11th international conference on learning representations (ICLR 2023)*.
- Deng, T., Huang, J., & Chen, Y. (2026). Metric learning based weighted linear discriminant analysis for imbalanced multi-label classification. *Applied Intelligence*, 56(1), 30.
- Dong, J., Jiang, J., Jiang, K., Li, J., & Zhang, Y. (2025). Fast and accurate gigapixel pathological image classification with hierarchical distillation multi-instance learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 30818–30828).
- Dong, S., Wang, P., & Abbas, K. (2021). A survey on deep learning and its applications. *Computer Science Review*, 40, 100379.
- Feng, L., Lv, J., Han, B., Xu, M., Niu, G., Geng, X., An, B., & Sugiyama, M. (2020). Provably consistent partial-label learning. *Advances in Neural Information Processing Systems*, 33, 10948–10960.
- Ilse, M., Tomczak, J., & Welling, M. (2018). Attention-based deep multiple instance learning. In *International conference on machine learning* (pp. 2127–2136).
- Kayed, M., Anter, A., & Mohamed, H. (2020). Classification of garments from fashion MNIST dataset using CNN lenet-5 architecture. In *2020 International conference on innovative trends in communication and computer engineering (ITCE)* (pp. 238–243). IEEE.
- LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (2002). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 2278–2324.
- Lin, J., Shen, Y., Wang, B., Lin, S., Li, K., & Cao, L. (2024). Weakly supervised open-vocabulary object detection. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 3404–3412). (vol. 38).
- Lowe, D. G. (2004). Distinctive image features from scale-invariant keypoints. *International Journal of Computer Vision*, 60, 91–110.
- Lv, J., Xu, M., Feng, L., Niu, G., Geng, X., & Sugiyama, M. (2020). Progressive identification of true labels for partial-label learning. In *International conference on machine learning* (pp. 6500–6510). PMLR.
- Maron, O., & Ratan, A. L. (1998). Multiple-instance learning for natural scene classification. In *Proceedings of the fifteenth international conference on machine learning* (pp. 341–349).
- Pan, Y., Chen, M., & Xu, W. (2026). An overlap-driven feature selection via volume-aware separability-consistency index for imbalanced classification. *Pattern Recognition*, 180, 113972.

- Qiao, C., Xu, N., Hu, Y., & Geng, X. (2025). Reduction-based pseudo-label generation for instance-dependent partial label learning. In *The thirty-ninth annual conference on neural information processing systems*.
- Settles, B., Craven, M., & Ray, S. (2007). Multiple-instance active learning. In *Proceedings of the 21st international conference on neural information processing systems* (pp. 1289–1296).
- Shao, D., Chen, R. J., Song, A. H., Runevic, J., Lu, M. Y., Ding, T., & Mahmood, F. (2025). Do multiple instance learning models transfer? In *International conference on machine learning* (pp. 54219–54238). PMLR.
- Shi, X., Xing, F., Xie, Y., Zhang, Z., Cui, L., & Yang, L. (2020). Loss-based attention for deep multiple instance learning. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 5742–5749). (vol. 34).
- Souriau, R., Martel, F., & Aksenova, T. (2026). Recursive sample weighted-n-way partial least squares for brain-computer interface decoder online learning with class imbalance. *Biomedical Signal Processing and Control*, 113, 109025.
- Tang, W., Yang, Y.-F., Wang, Z., Zhang, W., & Zhang, M.-L. (2024a). Multi-instance partial-label learning with margin adjustment. In *Advances in neural information processing systems 37, vancouver, canada*.
- Tang, W., Zhang, W., & Zhang, M.-L. (2023). Disambiguated attention embedding for multi-instance partial-label learning. *Advances in Neural Information Processing Systems*, 36, 56756–56771.
- Tang, W., Zhang, W., & Zhang, M.-L. (2024b). Exploiting conjugate label information for multi-instance partial-label learning. In *Proceedings of the thirty-third international joint conference on artificial intelligence* (pp. 4973–4981).
- Tang, W., Zhang, W., & Zhang, M.-L. (2024c). Multi-instance partial-label learning: Towards exploiting dual inexact supervision. *Science China Information Sciences*, 67(3), 132103.
- Taye, M. M. (2023). Understanding of machine learning with deep learning: architectures, workflow, applications and future directions. *Computers*, 12(5), 91.
- Wang, H., Yang, S., Lyu, G., Liu, W., Hu, T., Chen, K., Feng, S., & Chen, G. (2023). Deep partial multi-label learning with graph disambiguation. In *Proceedings of the thirty-second international joint conference on artificial intelligence* (pp. 4308–4316).
- Wang, W., Wu, D.-D., Wang, J., Niu, G., Zhang, M.-L., & Sugiyama, M. (2025). Realistic evaluation of deep partial-label learning algorithms. In *The thirteenth international conference on learning representations* (pp. 1–25).
- Wang, Y., Peng, J., Wang, H., & Wang, M. (2022). Progressive learning with multi-scale attention network for cross-domain vehicle re-identification. *Science China Information Sciences*, 65(6), 160103.
- Wei, X.-S., & Zhou, Z.-H. (2016). An empirical study on image bag generators for multi-instance learning. *Machine Learning*, 105, 155–198.
- Wen, H., Cui, J., Hang, H., Liu, J., Wang, Y., & Lin, Z. (2021). Leveraged weighted loss for partial label learning. In *International conference on machine learning* (pp. 11091–11100). PMLR.
- Witmer, A., & Bhanu, B. (2024). Iterative pseudo balancing for stem cell microscopy image classification. *Scientific Reports*, 14(1), 4489.
- Wu, D.-D., Wang, D.-B., & Zhang, M.-L. (2022). Revisiting consistency regularization for deep partial label learning. In *International conference on machine learning* (pp. 24212–24225). PMLR.
- Xiao, T., Sun, M., Xia, Y., Yang, H., Wang, Z., & Liu, P. (2026). Improving multi-instance learning with hierarchical attention and frequency-domain hard sample distillation. *Knowledge-Based Systems*, 347, 116286.
- Yildizer, E., Balci, A. M., Hassan, M., & Alhajj, R. (2012). Efficient content-based image retrieval using multiple support vector machines ensemble. *Expert Systems with Applications*, 39(3), 2385–2396.
- Zheng, T., Jiang, K., & Yao, H. (2024). Dynamic policy-driven adaptive multi-instance learning for whole slide image classification. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 8028–8037).
- Zhou, Z.-H. (2018). A brief introduction to weakly supervised learning. *National Science Review*, 5(1), 44–53.